

iMicroSeq Specification Testing Scenarios

Three water-based scenarios for testing the iMicroSeq contextual data specification |
Version 1.2

Item	Detail
Document	iMicroSeq_Worked-Example-Scenarios_v1.2.docx
Version	1.2
Status	Draft for testing round 1 - revised 2026-09-16
Specification	iMicroSeq contextual data specification v1.0 - 155 fields: 13 required, 48 recommended, 94 optional
Project	iMicroSeq, Genome Canada eDNA Surveillance Program / Centre for Infectious Disease Genomics and One Health, Simon Fraser University
Contact	iMicroSeq data structures team - Emma Griffiths, emma_griffiths@sfu.ca

How to use this document

Each scenario below is a short account of a piece of work shaped like the real thing - who sampled what, where, how, and what was done to it in the laboratory - but entirely invented. Your task in Part B of the testing protocol is to turn each narrative into a curated record set using the iMicroSeq template, working from the narrative alone. There are three, and we ask you to do all three: together they take an hour or two, and they are what makes your feedback in Part C comparable with everyone else's.

Only after you have curated a scenario should you open its answer key. The answer keys are in the Answer Keys/ folder, one workbook per scenario.

The three scenarios span the water types the iMicroSeq needs assessment identified as least well served by existing standards - municipal wastewater, drinking and source water, and industrially influenced water - and deliberately exercise the parts of the specification we are least confident about. Part B is a warm-up, not an exam: the point is to get you fluent enough in the template that Part C tells us something about the specification rather than about the tool.

	Scenario	Records	Answer key
S1	Municipal wastewater treatment plant influent	4	iMicroSeq_AnswerKey_S1_Municipal-Wastewater_v1.2.xlsx
S2	Source and drinking water - community-triggered monitoring	3	iMicroSeq_AnswerKey_S2_Drinking-Source-Water_v1.2.xlsx
S3	Landfill leachate and mining-influenced water - AMR metagenomics	3	iMicroSeq_AnswerKey_S3_Leachate-Mining-AMR_v1.2.xlsx

Everything in these scenarios is fictional

Every project, organisation, person, site name, city, coordinate, email address, identifier, accession and result in these scenarios and in the answer keys is invented. Names that look plausible - Kettle Harbour, Nickelford, Cedar Ridge Environmental Consulting - were made up for this exercise. Organisation names in the answer keys carry a '(fictional organisation)' tag, project names are prefixed '[FICTIONAL]', personal names carry '(fictional name)', email addresses and URLs use the reserved example.org domain, and coordinates are patterned decimals that point at no real facility. Nothing here describes a real site, programme or person, and nothing here needs to be treated as sensitive.

S1. Municipal wastewater treatment plant influent

Answer key	Records	Primary modules exercised
iMicroSeq_AnswerKey_S1_Municipal-Wastewater_v1.2.xlsx	4	Sample collection; Environmental conditions; Target and diagnostic measurements; Sequence information; Lineage/clade information

The scenario

A provincial public health laboratory runs a weekly SARS-CoV-2 and antimicrobial resistance monitoring programme at the Kettle Harbour Municipal Wastewater Treatment Plant, which serves an urban catchment of approximately 148,000 people.

Twenty-four hour flow-proportional composite samples of raw influent are collected by an ISCO autosampler at the influent pump station, starting at 08:00 each Tuesday and ending at 08:00 the following morning. The refrigerated composite is transported to the laboratory on the day of collection.

In the laboratory, 40 mL of the composite is concentrated by magnetic nanobead binding before nucleic acid extraction on a Maxwell RSC instrument. Pepper mild mottle virus (PMMoV) is quantified as a fecal strength normaliser. SARS-CoV-2 N1 and N2 targets are quantified by RT-qPCR, and the sul1 sulfonamide resistance gene is quantified by dPCR. Extracts positive for SARS-CoV-2 are taken forward for ARTIC v5.3.2 amplicon sequencing on an Illumina NextSeq 2000, with lineage assignment by Freyja.

Plant operators record daily total flow, influent pH, turbidity and total suspended solids. On 2026-03-10 the plant was carrying elevated flow following 22 mm of overnight rainfall. Sequencing is performed by the same laboratory that collects the samples.

Three consecutive weekly samples are included - 2026-03-03, 2026-03-10 and 2026-03-17 - plus one negative extraction control processed alongside the 2026-03-17 sample.

What this scenario is testing

You do not need to read this before curating - it is here so you know what we were probing when you come to compare against the answer key.

- Composite autosampler collection over a 24 hour window, including the duration value and unit pair.
- Wastewater treatment presampling activities.
- Catchment population recorded as both a measured value and a range.
- Flow, pH, turbidity and total suspended solids, each with its unit.
- Fecal strength normalisation using PMMoV.
- Three diagnostic target blocks used - two viral targets and one AMR gene.
- A negative extraction control curated as its own record, using null values and the experimental specimen role type field.

S2. Source and drinking water - community-triggered monitoring

Answer key	Records	Primary modules exercised
iMicroSeq_AnswerKey_S2_Drinking-Source-Water_v1.2.xlsx	3	Database identifiers; Sample collection; Environmental conditions; Sequence information

The scenario

A community environmental health team, working with a regional laboratory under a data sharing agreement, investigated a taste-and-odour complaint and a subsequent precautionary boil water advisory affecting a small groundwater system serving approximately 620 people.

Land guardians reported discoloured water at two households on 2026-06-02. Sampling took place on 2026-06-04 at three points: the source well head, the treated water leaving the chlorination unit, and the community treated water storage tank.

One litre was collected in a sterile bottle at each point as a grab sample. Samples were transported on ice and filtered on a 0.22 um membrane at the laboratory the same evening. Total coliforms and E. coli were enumerated by Colilert-18 (MPN per 100 mL). 16S rRNA V3-V4 amplicon sequencing was performed on an Illumina MiSeq, with taxonomic assignment against SILVA 138.2 using QIIME 2.

Precise well coordinates, the site name and the data steward's identity are held by the community and are not released outside the data sharing agreement.

What this scenario is testing

You do not need to read this before curating - it is here so you know what we were probing when you come to compare against the answer key.

- Well, drinking water treatment plant and water tank as environmental sites.
- Drinking water as an environmental material, with treated and untreated material properties.
- Grab sampling in bottles, with filtration recorded as specimen processing.
- Community-triggered environmental monitoring and land guardian observation as first-class terms.
- 16S rRNA amplicon sequencing with a named primer scheme and reference database.
- Data governance: using **Restricted Access** rather than blanks or approximations for location and personnel fields held under a data sharing agreement.

S3. Landfill leachate and mining-influenced water - AMR metagenomics

Answer key	Records	Primary modules exercised
iMicroSeq_AnswerKey_S3_Leachate-Mining-AMR_v1.2.xlsx	3	Sample collection; Environmental conditions; Sample processing; Target and diagnostic measurements

The scenario

An environmental consultancy, contracted by a provincial environment ministry, assessed antimicrobial resistance gene loads associated with two industrial land uses in the Nickelford area of northern Ontario: a legacy base-metal mine site and an adjacent municipal landfill.

Sampling took place on 2026-07-08. Three records were generated: a pooled sample of runoff from the mine tailings area, collected by torpedo passive water sampler deployed for seven days and pooled across three deployment points; landfill leachate collected by grab sampling from the leachate collection sump; and a downstream lake water sample serving as the receiving-environment comparator.

Runoff from the mine area had a pH of 4.1 and conductivity of 3.8 mS/cm. Leachate had a pH of 7.9 and conductivity of 14.2 mS/cm. Samples were filtered on site through 0.45 um syringe filters, extracted with the DNeasy PowerSoil Pro kit, and sequenced as shotgun metagenomes on an Illumina NextSeq 2000. AMR determinants were called with CARD RGI 6.0.3 against CARD 3.3.0, and taxonomic profiling used Kraken2 with the Standard PlusPF database.

No single target organism is under surveillance here, so `organism` is recorded at the level actually being characterised, and the diagnostic target blocks carry AMR gene targets rather than pathogens.

What this scenario is testing

You do not need to read this before curating - it is here so you know what we were probing when you come to compare against the answer key.

- Industrial land uses that existing wastewater standards cover poorly: mining and landfill.
- Leachate and runoff as environmental materials.
- Passive sampling with a torpedo sampler deployed over seven days, and pooled specimens.
- Conductivity and pH near the extremes of the plausible range - pH 4.1 acid rock drainage, 14.2 mS/cm leachate.
- AMR gene targets in the diagnostic target blocks in place of pathogen targets.
- A receiving-environment comparator record sharing a presampling activity with an upstream record.

Appendix: null values and when to use them

A blank cell tells a downstream user nothing. A null value tells them why the cell is empty, which is often as informative as the value would have been. All three answer keys use null values systematically; please do the same.

Null value	Use it when
Not Applicable [GENEPIO:0001619]	The field cannot apply to this record - for example catchment population at a mine tailings runoff site, or host fields for a water sample.
Missing [GENEPIO:0001618]	The value should exist but cannot be found, for an unknown reason.
Not Collected [GENEPIO:0001620]	The measurement was never taken, though it could have been.
Not Provided [GENEPIO:0001668]	The value exists and was collected, but is not being shared now. It may be shared later.
Restricted Access [GENEPIO:0001810]	The value exists but cannot be shared, for privacy, governance or data sovereignty reasons.

The distinction between **Not Applicable** and **Not Collected** is where testers most often diverge. If you find yourself hesitating, note the field on the feedback form - hesitation usually means the field definition needs sharpening.